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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION – March 2026

Fourth Semester – Discipline Specific Elective (DSE)

MG4DSEZGY201 - HEALTH, NUTRITION AND WELLNESS

Duration: 2 hrs **Maximum marks: 70**

Part A – Multiple Choice Questions

Answer all questions. Each question carries 1 mark.

1. The technique that blends breath focus with progressive muscle relaxation is: **[U] / [CO3]**
    - Answer: b). Body scan
2. Gout patients cannot have keto diet due to: **[K] / [CO2]**
    - Answer: c). Increased uric acid
3. Recommended sleep duration for adults is approximately: **[U] / [CO3]**
    - Answer: c). 7-9 hours
4. BMI is calculated by dividing weight by: **[U] / [CO1]**
    - Answer: b). Height²
5. Old age can be a time of _____ for individuals: **[K] / [CO5]**
    - Answer: d). Both B and C (Depression and Dementia)
6. Banana and orange are good sources of: **[U] / [CO1]**
    - Answer: a). Potassium
7. Any bodily movement produced by skeletal muscles that requires energy expenditure: **[K] / [CO4]**
    - Answer: d). Physical Activity
8. The main function of dietary antioxidants is to: **[U] / [CO1]**
    - Answer: c). Neutralize free radicals
9. Malnutrition due to nutritional abundance is termed as: **[K] / [CO2]**
    - Answer: a). Over nutrition
10. Which nutrient is essential for bone strength? **[K] / [CO1]**
    - Answer: a). Calcium

Part B – Short Answer Questions

Expected points for any 10 questions (3 marks each)

11. Mention any three physical health benefits of Yoga. [K] / [CO4]

Each point- 1 mark

Key Points: Improved flexibility and balance; increased muscle strength and tone; enhanced cardiovascular and circulatory health.

12. List three effects of regular exercise on the respiratory system. [U] / [CO4]

Each point- 1 mark

Key Points: Increased vital capacity; improved efficiency of gas exchange in the lungs; strengthened respiratory muscles (like the diaphragm).

13. Name any two cardiac problems related to nutritional imbalance. [K] / [CO2]

Any two cardiac problems: 1.5 marks each

Key Points: Hypertension (high blood pressure) and Atherosclerosis (linked to high saturated fat/cholesterol intake).

14. What is sarcopenia, and how is it related to ageing? [U] / [CO5]

If definition and key points are correct give full marks

Key Points: Sarcopenia is the involuntary loss of skeletal muscle mass and strength; it is a natural part of the ageing process but can be accelerated by poor nutrition and inactivity.

15. List three common symptoms of lactose intolerance. [U] / [CO2]

Each point- 1 mark

Key Points: Abdominal bloating; diarrhea; gas (flatulence) and stomach cramps after consuming dairy.

16. What is the role of dietary fiber in obesity management? [U] / [CO2]

Any 3 briefly explained points- 1 mark each

Key Points: Promotes satiety (feeling of fullness), which reduces overall calorie intake; slows down the digestion and absorption of nutrients.

17. Differentiate between simple proteins and conjugated proteins. [K] / [CO1]

Definitions- 2 marks, examples-1 mark

Key Points: Simple proteins yield only amino acids upon hydrolysis (e.g., albumin); conjugated proteins consist of amino acids combined with a non-protein group (e.g., hemoglobin with heme).

18. Explain the threat of fake paneer and other milk products in Indian market. [U] / [CO2]

Liberal valuation

Any relevant 3 points- full marks

Key Points: Use of harmful adulterants like urea, detergents, and vegetable oils to mimic fat content; poses severe long-term health risks like kidney damage and digestive disorders.

19. Mention two units used to express BMR. [K] / [CO1]

Liberal valuation

Give full marks if definition/ full form of BMR is correct with relevant 3 points

Or

Any relevant 3 points & one of the units [kcal/day (calories per day) and kJ/day (kilojoules per day)] ***is correct.***

20. What are structural carbohydrates? Give examples. [U] / [CO1]

Definition with examples- 2 marks; biological importance – 1mark

Key Points: Carbohydrates that provide physical form and support to organisms; examples include **Cellulose** (plant cell walls) and **Chitin** (exoskeletons).

21. How do fats act as body insulators? [U] / [CO1]

Liberal valuation- any 2 relevant points give full marks

Key Points: Adipose tissue stored under the skin (subcutaneous fat) has low thermal conductivity, which helps retain body heat and maintain core temperature.

22. Differentiate hypovitaminosis and avitaminosis. [U] / [CO2]

Liberal valuation **Definitions- 1.5 marks each**

Key Points: Hypovitaminosis is a partial deficiency or low level of a vitamin; Avitaminosis is a severe or total lack of a specific vitamin leading to clinical disease (e.g., Scurvy).

Part C – Short Essay Questions

Explanatory summaries for any 6 questions (5 marks each).

- 23. Explain how poor nutrition during adolescence affects adult life. [U] / [CO5]

Any 5 relevant points with explanation- 1 mark each

  - Adolescence is a critical growth phase. Poor nutrition can lead to stunted physical growth, reduced bone density (increasing osteoporosis risk later), and a higher predisposition to chronic diseases like obesity, diabetes, and heart disease in adulthood.

- 24. Explain why older adults are more susceptible to nutritional problems. [U] / [CO5]

Liberal valuation

Any 5 relevant points with explanation- 1 mark each

  - Susceptibility is due to physiological changes (reduced metabolism, impaired digestion), sensory decline (loss of taste/smell), dental issues, and social factors like isolation or limited mobility that affect food access.

- 25. Comment on the importance of nutrition during third trimester of pregnancy. [U] / [CO5]

Importance of nutrition in pregnancy- 1 marks; Importance of third trimester- 1 mark; Additional nutritional requirements during third trimester and its relevance- 3 marks

- The third trimester involves rapid foetal brain development and weight gain. Adequate intake of calcium (for bones), iron (for blood volume), and DHA (for brain/eyes) is vital to prevent low birth weight and maternal depletion.

- 26. Explain the role of Vitamin D in the human body. Mention its sources and the effects of deficiency and excess intake. [U] / [CO1]

  Sources- 1 mark; Importance- 2 marks; deficiency diseases- 1 mark; excess intake- 1 mark

  - **Role:** Essential for calcium absorption and bone mineralization. **Sources:** Sunlight, fatty fish, egg yolks. **Deficiency:** Rickets (children), Osteomalacia (adults). **Excess:** Hypercalcemia (calcium buildup in blood/kidneys).

- 27. Comment on various antibodies in colostrum. [U] / [CO5]

  Liberal valuation

  Any 5 relevant points related to colostrum and its importance- 1 mark each

  - Colostrum (the first breast milk) is rich in **Secretory IgA**, which provides passive immunity by coating the infant's gut to prevent pathogens from entering. It also contains **IgG** and **IgM** to support the developing immune system.

- 28. Define childhood and mention the age groups included under childhood nutrition. [K] / [CO5]

  Definition- 1 mark; mention each of 4 groups with brief explanation- 1 mark each

  - Childhood is the developmental period from infancy to the onset of puberty. Age groups include **Infancy** (0-1 year), **Toddlerhood** (1-3 years), **Preschool** (3-5 years), and **School-age** (6-12 years).

- 29. An elderly patient with severe vomiting is admitted to the hospital. Suggest an adequate hydration strategy with reasons. [U] / [CO5]

  Liberal valuation- Give full marks if logical points related to elderly hydration is given

  - **Strategy:** Immediate Intravenous (IV) fluids (like Normal Saline or Ringer's Lactate) followed by gradual Oral Rehydration Salts (ORS). **Reason:** Severe vomiting causes rapid electrolyte loss; elderly patients have lower total body water and are at high risk for kidney failure if not rehydrated quickly.

- 30. Explain "Nutrition is a powerful tool for controlling diabetes mellitus." [U] / [CO2]

  Importance of nutrition in controlling Diabetes- 2 marks; various food examples for diabetes patients- 2 marks; logical explanations- 1 mark.

  - Medical Nutrition Therapy (MNT) focuses on controlling blood glucose levels through low-glycemic index foods, high fiber, and portion control. Proper nutrition reduces reliance on medication and prevents complications like neuropathy and retinopathy.